

Jahnvi Mandhadapu

Tampa, Florida

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SUMMARY

Bioinformatics and biotechnology- trained professional with hands-on experience in cell culture and bioreactor data analysis. Skilled in implementing analytical tools (ELISA, microscopy), maintaining GMP-compliant databases, and troubleshooting production processes. Combines computational biology expertise (Python/R, data visualization) with wet-lab proficiency (RNA-seq, PCR, sterile techniques) to optimize bioproduction workflows. Adept at SOP development, trend analysis, and cross-functional collaboration in regulated environments.

EDUCATION

UNIVERSITY OF SOUTH FLORIDA, Tampa, FL.

Aug 2022 - May 2024

MASTER OF SCIENCE, BIOINFORMATICS & COMPUTATIONAL BIOLOGY

• **Coursework:** R programming, Directed research, Data Analysis, Biostatistics

VIGNAN'S DEEMED TO BE UNIVERSITY, India.

Jul 2017 - May 2021

BACHELOR OF TECHNOLOGY, BIOINFORMATICS

• **Coursework:** Bioinformatics, Microbiology, Molecular Biology, Genomics, Proteomics, Genetics.

EXPERIENCE

MOFFITT CANCER CENTER

Nov 2024 – July 2025 TAMPA, FL

Research Student Assistant - Predictive Modeling of Drug Response Based on Gene Expression in Cancer Cell Lines

- Retrieved and integrated large-scale gene expression and drug response (gemcitabine) datasets from global databases including GDSC, CTRP, and CCLE.
- Performed data preprocessing and normalization using R and bioinformatics libraries to ensure compatibility across sources.
- Conducted correlation analysis between gene expression profiles and drug sensitivity metrics across multiple datasets (GDSC, CCLE).
- Developed and evaluated predictive models to assess the relationship between gene expression patterns and gemcitabine response, using statistical and machine learning methods in R.
- Written a report and research article to visualize the summary of the project.
- Aimed to identify consistent biomarkers of drug response across different data sources to support precision oncology research.

QSTATIX PVT LTD

Mar 2021-Jul 2021 INDIA

Drug Discovery Intern— Drug discovery approach for Type 2 diabetes

- Actively engaged in an internship involving drug discovery for Type 2 diabetes inhibitors.
- Collaborate with peers in the bioinformatics team to create operational pipelines preceding customized analysis workflows.
- Conducted in vitro and ex vivo studies to facilitate drug discovery independently.
- Recorded all details pertaining to computational environment parameters, instructions, and standard operating procedures (SOPs) for both existing and newly implemented systems.
- Collaborated with 'wet-lab' colleagues to propose, develop, execute, and interpret methodologies for studying multi-omic datasets, focusing on epigenetic platforms such as ChIP-seq, CUT&TAG, and ATAC-seq.
- Focused on the identification of novel compounds with the potential to inhibit Type 2 diabetes.
- Utilized molecular docking and virtual screening methodologies as key research approaches.
- Made significant contributions to the project by applying these techniques effectively.
- Identified and recommended potential new compounds for further investigation

Digital Intern

- Undertook an online digital internship at IIIT Hyderabad (SERC) focused on web development
- Acquired practical skills and experience in diverse areas of web development during the internship
- Executed a notable project that involved the practical application of HTML and CSS
- Proficiently employed GitHub as a collaborative platform for uploading and managing project-related documents.

PROJECTS

ANALYSIS OF INTRINSIC DISORDER STATUS OF RADIOTROPHIC FUNGI & NEUROTROPIC FACTORS

- Analyzed and integrated potential functional implications of IDPs in both Radiotrophic and neurotrophic factors.
- Conducted a wide range analysis by using RIDAO, D2P2, FUZDROP, ALPHAFOLD, and STRING.
- Written a report and reviewed articles and created visual figures and summarized biological implications.

EXTRACTION & PARTIAL PURIFICATION OF HYDROGEN PEROXIDE OXIDOREDUCTASE FROM *MACLURA POMIFERA*

- Collected fresh Osage orange fruit, flash-froze in liquid nitrogen, and ground into powder.
- Homogenized tissue in phosphate buffer (pH 7.0, 1 mM EDTA, 1% PVP), centrifuged (10,000 × g, 20 min).
- Precipitated proteins at 30–60% saturation, pelleted via centrifugation.
- Desalted using 10 kDa MWCO tubing against phosphate buffer (pH 7.0).
- Quantified H₂O₂ degradation at 240 nm ($\epsilon = 43.6 \text{ M}^{-1} \text{ cm}^{-1}$) to measure oxidoreductase activity.
- Determined concentration via Bradford assay (595 nm).
- Isolated partially purified enzyme; confirmed peroxidase/catalase-like activity.

TRANSCRIPTOMIC DATA PIPELINE FOR DIFFERENTIAL EXPRESSION

- Designed full RNA-seq analysis pipeline including alignment, quantification, normalization, and visualization
- Integrated statistical models to assess biological significance and pathway impact.

R PACKAGE

- Created a package and used different functions to preprocess the data to support analysis and data reporting.
- Used RMarkdown and the visualization process.

SKILLS

- **Programming Languages:** Python, R Shiny, C
- **Data Analysis:** SQL, Power BI, Tableau, Statistical Analysis, Annotation techniques
- **Bioinformatics Tools:** NCBI, PDB, ENCODE, Uniport, Ridao, String, Fuzdrop, D2P2, AWS, Data Visualization Techniques
- **Cloud Platforms:** Google Cloud Platform (GCP)
- **Microsoft Applications:** Word, Excel, Powerpoint, Linux
- **Laboratory Skills:** RNA Seq, Gene Expression, Genome Assembly, Metabolomics, PCR, NGS, Protein Purification, Post-translational Modification, Clustering, Molecular Docking, ELISA, Mass Spectrometry, Flourescent microscopy, Virtual Screening
- **Clinical Data Integration:** Familiar with clinical variables, demographic stratification, and survival analysis.
- **Soft Skills:** Exceptional communication, Team Collaboration, Analytical review, Research writing, Scientific Documentation, SOPs, Problem Solving